

MUGHEES HIADER

Aspiring AI Engineer | RAG & LLM Applications | Intelligent Software Systems

i243181@isb.nu.edu.pk | Pakistan | FAST-NUCES Islamabad

PROFESSIONAL SUMMARY

Ambitious Software Engineering student focused on AI engineering and practical intelligent systems. Building across RAG, LLM applications, image-based AI, local model deployment, databases, algorithms, concurrency, and low-level programming.

EDUCATION

BS Software Engineering Expected 2028

FAST - National University of Computer and Emerging Sciences | Islamabad

Coursework: Data Structures, OOP, Operating Systems, Database Systems, Software Design & Architecture, Assembly Language Programming, Civics.

Real-life implementation: Applied theory through hands-on systems, graphics, web, AI, database, and full-stack projects.

TECHNICAL SKILLS

AI Engineering: RAG, Foundation Models & LLMs, Prompt Engineering, LLM APIs, Local LLM Deployment, Model Inference, AI Application Architecture, Basic Model Evaluation

AI Tools: Ollama, Grok API, local open-source LLMs, embeddings (learning), vector databases (learning)

Programming: Python, TypeScript, C, C++, Java, SQL, x86 Assembly, Bash / Command Line

Development & Systems: Docker, Git/GitHub, Linux/Ubuntu, VS Code, SFML/GLUT, DSA, OOP, POSIX Threads, Semaphores, IPC, ACID, RBAC

AI & INTELLIGENT SYSTEMS PROJECTS

FixFlow AI - RAG-Based Software Debugging Assistant | [RAG](#) | [Official Technical Documentation](#)

- Developing an AI debugging assistant that retrieves official documentation to diagnose errors across modern software technologies.
- Uses grounded, source-aware retrieval so troubleshooting guidance stays tied to authoritative technical material.

LeafScan AI - Plant Disease Analysis | [TypeScript](#) | [Grok API](#)

- Built an AI application that analyzes leaf images to identify likely plant diseases.
- Integrated Grok API workflows to return structured treatment guidance in a practical user application.

Local LLM Deployment with Ollama | [Ollama](#) | [Linux/Ubuntu](#) | [Local Models](#)

- Installed, configured, and tested open-source language models locally for offline inference and experimentation.
- Worked with model storage, local ports, lifecycle management, and application-to-model runtime communication.

DATA, WEB & BUSINESS SYSTEMS

Smart Disaster Response MIS | [Full-Stack Web App](#) | [SQL](#) | [RBAC](#)

- Built a disaster-response MIS with normalized relational data, ACID transactions, triggers, and five-role RBAC.
- Implemented indexing, dashboards, reporting, and comparative query-performance analysis.

DATA, WEB & BUSINESS SYSTEMS - CONTINUED

Point-of-Sale (POS) System | [TypeScript](#)

- Built a full POS system covering sales, products, inventory, transaction records, and sales history.
- Added reporting, backup, and recovery capabilities for dependable store operations.

Traffic Company Management System | [Java](#)

- Built a Java application for customer complaint tracking, reviews, and related case information.
- Structured follow-up using review and justification records for clearer issue handling.

ALGORITHMS, SYSTEMS & CONCURRENCY

OceanRoute Nav - Maritime Navigation Optimizer | [C++](#) | [Custom Data Structures](#) | [SFML/GLUT](#)

- Built a zero-STL maritime optimizer using custom graph, linked list, priority queue, and stack structures.
- Implemented Dijkstra routing, FIFO docking queues, multi-leg routes, subgraph filtering, and route preferences.

Traffic Intersection Simulator | [C](#) | [POSIX Threads](#) | [Semaphores](#) | [IPC Pipes](#)

- Simulated two concurrent intersections using pthreads, semaphores, and fork-based IPC.
- Implemented ambulance priority, bounded parking, race-condition prevention, coordinated flow, and SIGINT shutdown.

SOFTWARE ENGINEERING & QUALITY

CCTNS-Inspired Crime & Criminal Tracking System | [Full-Stack](#) | [SonarQube](#) | [Software Testing](#)

- Developed an academic crime and case-management system with complaint registration, investigation, search, RBAC, and audit tracking.
- Performed functional/system testing and SonarQube analysis for bugs, vulnerabilities, code smells, and maintainability issues.

DEVELOPMENT & DEVOPS

Containerized Application Environment | [Docker](#) | [Containers](#) | [Networking](#)

- Created reproducible Docker environments using images and containers.
- Worked with container networking, port mapping, volumes, and deployment-oriented workflows.

GRAPHICS & GAME DEVELOPMENT

UNO Card Game | [C++](#) | [SFML](#)

- Built a graphical UNO game with deck management, turn logic, special-card effects, player interaction, and rendering.
- Applied object-oriented programming and real-time input handling for complete game flow.

Brick Breaker Game | [x86 Assembly Language](#)

- Created a playable x86 Assembly game with low-level graphics, keyboard input, collision detection, and score tracking.
- Applied direct program-flow control and hardware-near concepts to implement interactive gameplay.

COMMUNITY-BASED WORK

Islamabad Wildlife Park - Preservation Awareness Initiative | [Civics Course Project](#) | [FAST-NUCES](#)

- Completed field-based community work focused on wildlife preservation and public awareness.
- Documented species at risk, researched conservation threats, and prepared practical policy recommendations.